

MachineEcho

Contactless Machine Anomaly Detection on Snapdragon PCs

1. Problem Statement

Everyday machines — ceiling fans, water pumps, washing machines, compressors, and small motors — often develop subtle abnormal acoustic and vibrational patterns long before catastrophic failure. Detecting these early usually requires expensive specialized hardware sensors, dedicated technicians, or periodic teardown inspections. Consequently, predictive maintenance remains largely inaccessible for resource-constrained households, small workshops, and local businesses.

2. Proposed Solution

MachineEcho turns an ordinary Snapdragon-powered HP laptop into a contactless machine-health monitoring device using its built-in microphone and on-device AI. The system operates in three simple phases:

- **Learn Baseline:** Records 30–60 seconds of a machine's normal acoustic signature to train a custom baseline.
- **Real-Time Monitoring:** Continuously evaluates live audio windows (0.96s) accelerated via Snapdragon Hexagon NPU.
- **Interpretable Warning:** Outputs real-time status: *Normal*, *Anomaly Detected*, or *High-Risk Anomaly*.

3. Technical Architecture & Innovation

- **Audio Pipeline:** Built-in Mic (16 kHz mono) → Log-Mel Spectrogram (96×64 patch).
- **Feature Extractor:** YAMNet ONNX model (Qualcomm AI Hub) producing 1024-dim embeddings.
- **Anomaly Model:** One-Class Isolation Forest trained exclusively on normal operation data.
- **Calibrated Scoring:** Z-score normalization converts raw scores to intuitive [0, 1] anomaly metric.
- **Zero Cold-Start:** Machine-agnostic architecture capable of inspecting novel appliances without retrain datasets.

4. Why Snapdragon?

- **Dedicated NPU Execution:** Uses Qualcomm's Hexagon NPU for sub-millisecond inference, preventing CPU thermal throttling.
- **100% On-Device & Private:** All acoustic analysis remains local on the PC. Zero audio streaming to cloud servers.
- **Qualcomm AI Hub Integration:** Pre-optimized YAMNet model profiles target Snapdragon X Elite / X Plus architecture.
- **Low Power Consumption:** Enables hours of continuous contactless monitoring on battery power.

5. Empirical Verification Results

Condition	Mean Anomaly Score	Status	Detection Recall
Normal Operation	0.1367	NORMAL [OK]	Baseline Match
Abnormal Operation	0.6195	ANOMALY DETECTED [!!]	28 / 30 Windows (93.3%)